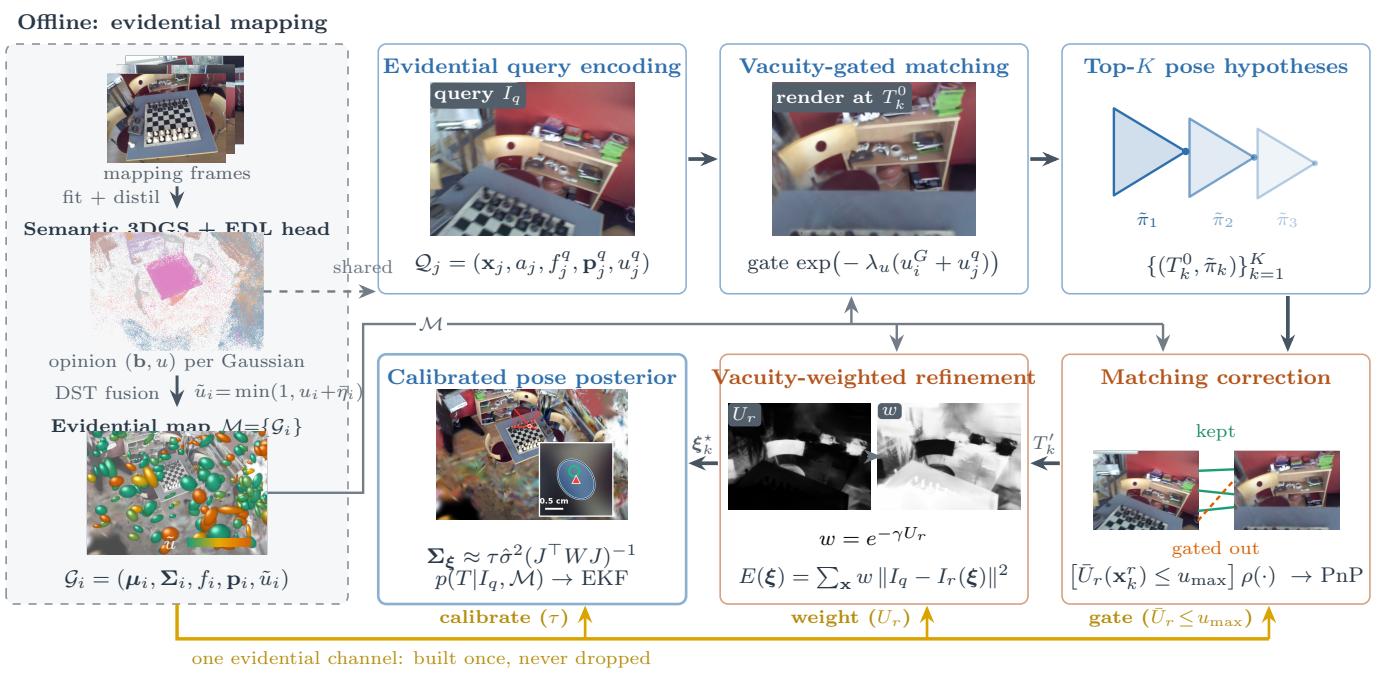

Offline: evidential mapping
mapping frames
fit + distil
Semantic 3DGS + EDL head
shared
opinion $(\mathbf{b}, u)$ per Gaussian
DST fusion
$\tilde{u}_i = \min(1, u_i + \bar{\eta}_i)$
Evidential map $\mathcal{M} = \{\mathcal{G}_i\}$
$\mathcal{G}_i = (\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, f_i, \mathbf{p}_i, \tilde{u}_i)$
Evidential query encoding
query I_q
$\mathcal{Q}_j = (\mathbf{x}_j, a_j, f_j^q, \mathbf{p}_j^q, u_j^q)$
Vacuity-gated matching
render at T_k^0
gate $\exp\big(-\lambda_u(u_i^G + u_j^q)\big)$
Top-K pose hypotheses
$\tilde{\pi}_1$
$\tilde{\pi}_2$
$\tilde{\pi}_3$
$\{(T_k^0, \tilde{\pi}_k)\}_{k=1}^K$
$\mathcal{M}$
Calibrated pose posterior
0.5 cm
$\boldsymbol{\Sigma}_{\boldsymbol{\xi}} \approx \tau\hat{\sigma}^2 (J^\top W J)^{-1}$
$p(T|I_q, \mathcal{M}) \to$ EKF
Vacuity-weighted refinement
U_r
w
$w = e^{-\gamma U_r}$
$E(\boldsymbol{\xi}) = \sum_{\mathbf{x}} w \, \|I_q - I_r(\boldsymbol{\xi})\|^2$
Matching correction
kept
gated out
$\big[\bar{U}_r(\mathbf{x}_k^r) \le u_{\max}\big]\, \rho(\cdot) \;\to$ PnP
$\xi_k^\star$
T_k'
calibrate (τ)
weight (U_r)
gate ($\bar{U}_r \le u_{\max}$)
one evidential channel: built once, never dropped